MFE 293 Independent Study Syllabus

Student:	Ronald Liu	Term:	Summer 2026
Faculty Supervisor:	Prof. Thibaut Mastrolia (proposed)	Duration:	8 weeks (June 1 to July 26, 2026)
		Units:	2

Project in One Paragraph

Alpha-GPT 3.0 uses multiple LLM agents to debate trading ideas and turn them into formulaic alpha factors, which then seed a genetic-programming search. The current results are preliminary: a narrow 500-stock sample, weak baselines, no ablations on the framework's own design choices. This study fixes those gaps to answer one question: **does agent debate actually help, and if so, what part of it helps?**

Week-by-Week Plan

Week	Dates	Focus	Deliverable
1	Jun 1–Jun 7	**Get the data right.** Pull a broader stock universe and richer features from WRDS so results don't depend on a small sample. A larger feature set also plays to the framework's strength, since LLM agents can use economic intuition to navigate it while brute-force symbolic search scales exponentially.	Clean data panel; one-page data-quality report.
2	Jun 8–Jun 14	**Re-run the existing pipeline on the new data.** Confirm the harness still works end-to-end and re-measure all baselines on the bigger panel.	Baseline results table on the expanded universe. First check-in.
3	Jun 15–Jun 21	**Make the backtests trustworthy.** Run each configuration many times with different random seeds and report confidence intervals instead of single numbers.	Baseline table with error bars.
4	Jun 22–Jun 28	**Make the backtests realistic.** Add transaction costs, measure turnover, and check whether the signal works in different market regimes (calm vs. volatile).	Cost-adjusted Sharpe table; regime breakdown. Second check-in.
5	Jun 29–Jul 5	**Does adding more agents help?** Re-run the system with 1, 3, and 5 agents (adding quality and low-volatility roles for the larger setup) and see how performance scales.	Comparison across agent counts.
6	Jul 6–Jul 12	**Does agent diversity help?** Hold the count fixed at 3 and compare three identical agents against three with different priors (momentum, mean-reversion, fundamental).	Comparison across role compositions. Third check-in.
7	Jul 13–Jul 19	**Why does it work or fail?** Look inside the debate logs to see where invalid formulas come from, which agents revise most, and which ideas survive to the final output.	Failure-mode catalog; per-agent revision statistics.
8	Jul 20–Jul 26	**Write it up.** Draft the paper, finalize plots, clean up the codebase for release.	Workshop paper draft; reproducible codebase. Final check-in.

Supervision Cadence

Bi-weekly 30-minute meetings (Weeks 2, 4, 6, 8). Email updates the other weeks.